

1 — Installing SDOM (Step-by-Step Tutorial)

This tutorial walks you through installing the **Storage Deployment Optimization Model (SDOM)** on a Windows machine. It is based on the official SDOM documentation:

- SDOM Documentation home: <https://natlabrockies.github.io/SDOM/index.html>
- Developer / environment setup guide: https://natlabrockies.github.io/SDOM/sdom_Developers_guide.html
- SDOM GitHub repository: <https://github.com/NatLabRockies/SDOM>

SDOM is an open-source, Python/Pyomo-based, high-resolution capacity-expansion framework developed by the National Lab of the Rockies (NLR). See the [SDOM home page](#) for an overview.

Table of Contents

1. [Prerequisites](#)
 2. [Install Python](#)
 3. [Install VS Code and recommended extensions](#)
 4. [Install uv \(Python environment manager\)](#)
 5. [Create a virtual environment](#)
 6. [Install SDOM](#)
 - [Option A — Install from PyPI \(end users\)](#)
 - [Option B — Install from source \(developers\)](#)
 7. [Install a solver \(CBC / HiGHS\)](#)
 8. [Verify the installation](#)
 9. [Quick Start example](#)
 10. [Troubleshooting](#)
 11. [References](#)
-

1. Prerequisites

Before starting, make sure you have:

- A Windows, macOS or Linux machine with administrator rights.
- An internet connection (to download Python, [uv](#), and SDOM).
- ~2 GB of free disk space (Python + virtual env + solver).

Reference: [SDOM Installation — System Setup and Prerequisites](#)

2. Install Python

1. Download Python (3.10 or newer is recommended) from the official site: <https://www.python.org/downloads/>
2. Run the installer. **Important:** on the first installer screen, check the box "**Add Python to PATH**".

3. Confirm Python is on your **PATH**. If not, follow this guide to add it: <https://realpython.com/add-python-to-path/>
4. Open a new PowerShell window and verify:

```
python --version  
pip --version
```

You should see Python and pip versions printed.

3. Install VS Code and recommended extensions

SDOM docs recommend [Visual Studio Code](#) as the IDE.

1. Download and install VS Code: <https://code.visualstudio.com/>
2. Install these recommended extensions from the VS Code Marketplace:
 - **Python** (Microsoft)
 - **edit CSV** — to edit SDOM input CSVs directly in VS Code:
<https://marketplace.visualstudio.com/items?itemName=janisdd.vscode-edit-csv>
 - **vscode-pdf** — to view PDFs inside VS Code: <https://marketplace.visualstudio.com/items?itemName=tomoki1207.pdf>

4. Install **uv** (Python environment manager)

SDOM uses **uv** to manage virtual environments and dependencies.

Open PowerShell and run:

```
pip install uv
```

Verify:

```
uv --version
```

Reference: [Install uv](#)

5. Create a virtual environment

Choose (or create) a working folder for your SDOM project, **cd** into it, and create a virtual environment named **.venv**:

```
# Move to your project folder
cd C:\Users\<your-user>\repos\<your-project>

# Create the virtual environment
uv venv .venv

# Activate it (Windows PowerShell)
.venv\Scripts\Activate.ps1
```

For macOS / Linux:

```
source .venv/bin/activate
```

When activation succeeds the prompt is prefixed with `(.venv)`.

If PowerShell blocks script activation, run once (as Administrator): `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`

6. Install SDOM

You have two ways to install SDOM. Pick the one that matches your use case.

With the `.venv` activated:

```
uv pip install sdom
```

7. Verify the installation

With the virtual environment activated:

```
python -c "import sdom; print(sdom.__version__)"
```

You should see a version number printed (no `ImportError`).

8. Quick Start example

A minimal SDOM run looks like this (from the [official Quick Start](#)):

```
from sdom import (
    load_data,
```

```

        initialize_model,
        run_solver,
        get_default_solver_config_dict,
        export_results,
    )

# Load input data
data = load_data("./Data/your_scenario/")

# Initialize model (8760 hours = full year)
model = initialize_model(data, n_hours=8760)

# Configure solver (CBC example)
solver_config = get_default_solver_config_dict(
    solver_name="cbc",
    executable_path="./Solver/bin/cbc.exe",
)

# Solve
results = run_solver(model, solver_config)

if results.is_optimal:
    print(f"Total Cost: ${results.total_cost:,.2f}")
    print(f"Wind Capacity: {results.total_cap_wind:.2f} MW")
    print(f"Solar Capacity: {results.total_cap_pv:.2f} MW")
    export_results(results, case="scenario_1", output_dir="./results/")

```

For details on inputs and outputs, see:

- [SDOM Input Data](#)
- [Running SDOM and Understanding Outputs](#)

9. Troubleshooting

- **uv is not recognized** — Re-open PowerShell after `pip install uv`, or check your `PATH`.
 - **Activation script is blocked** — run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` in PowerShell.
 - **ImportError: sdm** — make sure the `.venv` is **activated** before running Python; the prompt should start with `(.venv)`.
 - **Hardlink warning during uv pip install** — harmless; you can ignore it or set `set UV_LINK_MODE=copy`.
 - **Solver not found** — pass an explicit `executable_path` to `get_default_solver_config_dict(...)` pointing to your CBC/HiGHS binary.
 - See also: [Troubleshooting](#).
-

11. References

- [SDOM Documentation \(home\)](#)
- [Installation section](#)

- [Developer / environment setup guide](#)
- [SDOM GitHub repository](#)
- [User Guide — Introduction](#)
- [User Guide — Input Data](#)
- [User Guide — Running SDOM and Outputs](#)
- [API Reference](#)
- [uv on PyPI](#)
- [Python downloads](#)
- [VS Code](#)